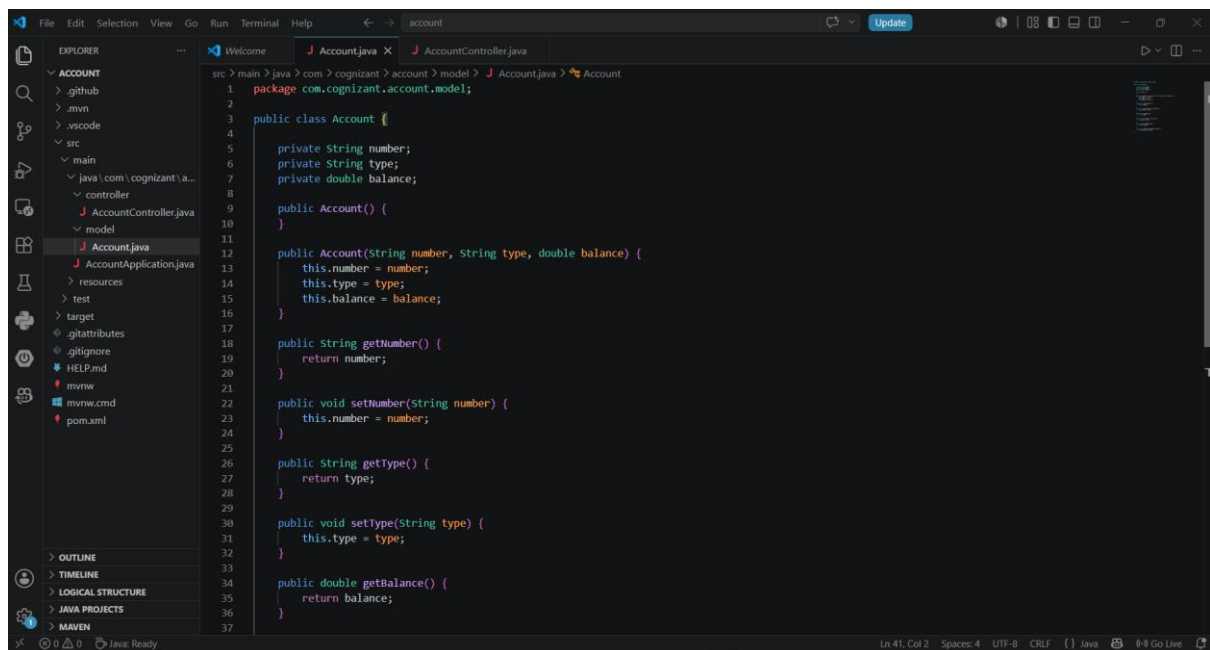


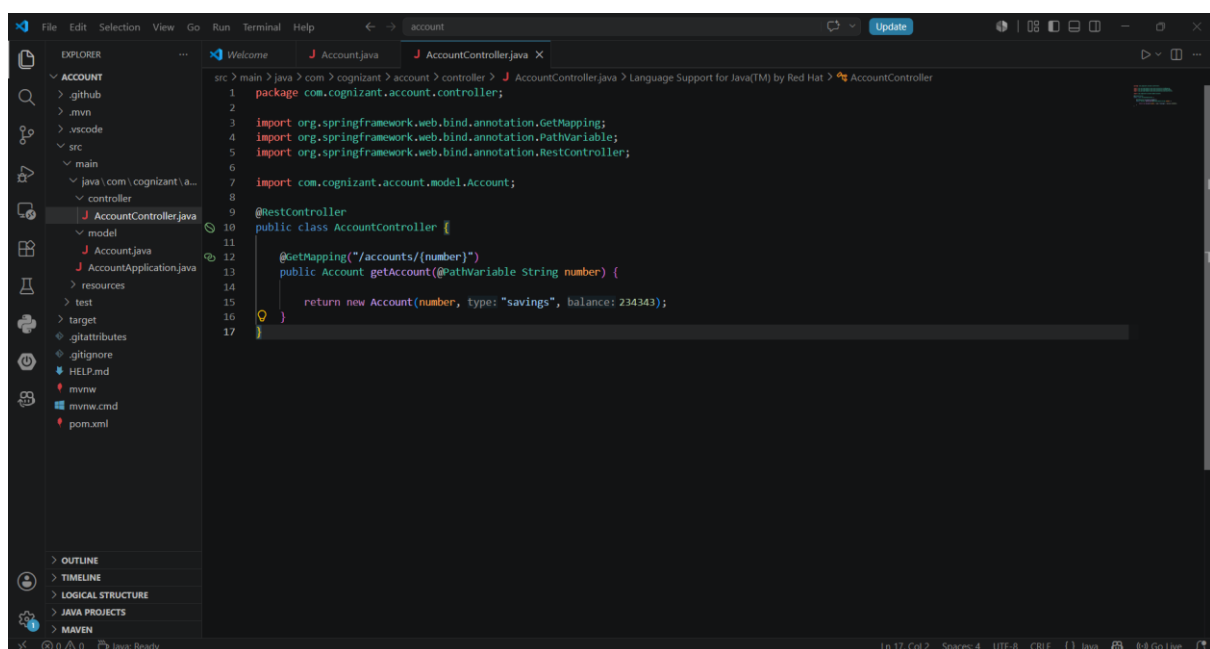
Creating Microservices for account and loan

In this hands on exercises, we will create two microservices for a bank. One microservice for handling accounts and one for handling loans. Each microservice will be a specific independent Spring RESTful Webservice maven project having it's own pom.xml. The only difference is that, instead of having both account and loan as a single application, it is split into two different applications. These webservices will be a simple service without any backend connectivity.

Account.java

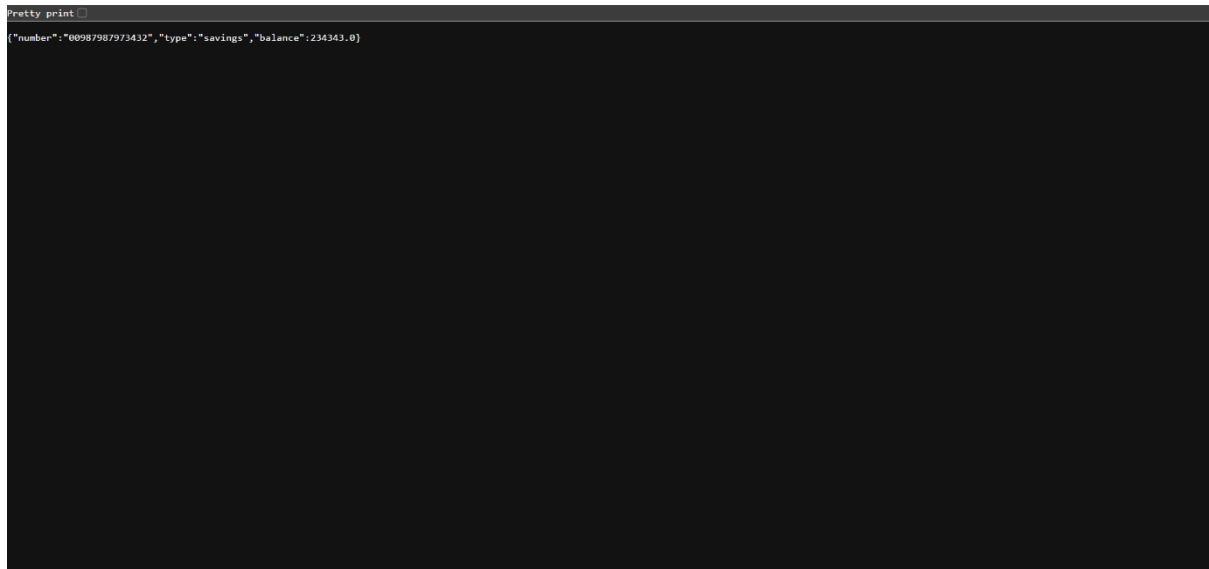
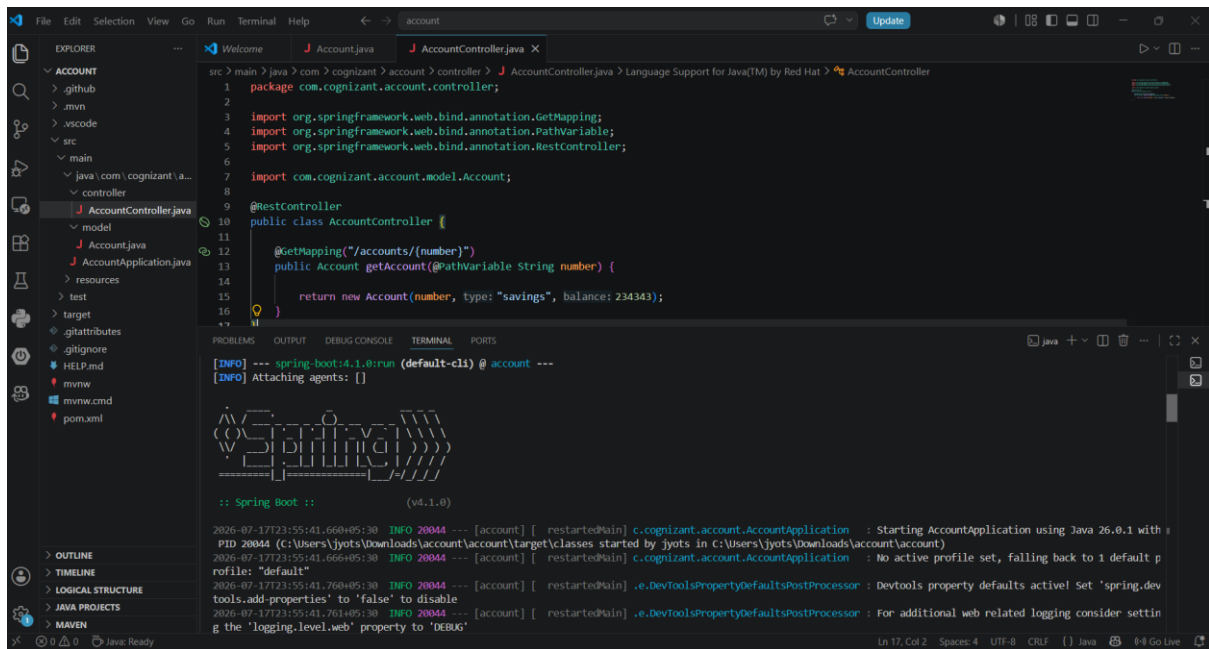


```
src > main > java > com > cognizant > account > model > Account.java > Account
1 package com.cognizant.account.model;
2
3 public class Account {
4
5     private String number;
6     private String type;
7     private double balance;
8
9     public Account() {
10
11     }
12
13     public Account(String number, String type, double balance) {
14         this.number = number;
15         this.type = type;
16         this.balance = balance;
17     }
18
19     public String getNumber() {
20         return number;
21     }
22
23     public void setNumber(String number) {
24         this.number = number;
25     }
26
27     public String getType() {
28         return type;
29     }
30
31     public void setType(String type) {
32         this.type = type;
33     }
34
35     public double getBalance() {
36         return balance;
37     }
38 }
```



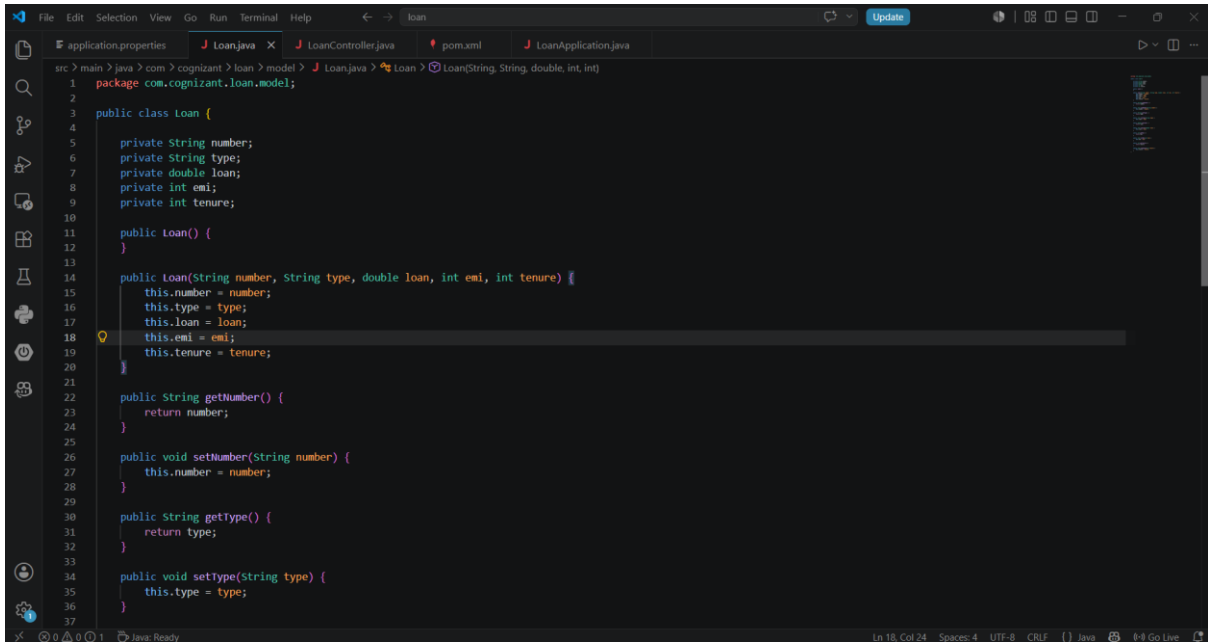
```
src > main > java > com > cognizant > account > controller > AccountController.java > Language Support for Java(TM) by Red Hat > AccountController
1 package com.cognizant.account.controller;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.PathVariable;
5 import org.springframework.web.bind.annotation.RestController;
6
7 import com.cognizant.account.model.Account;
8
9 @RestController
10 public class AccountController {
11
12     @GetMapping("/accounts/{number}")
13     public Account getAccount(@PathVariable String number) {
14
15         return new Account(number, type: "savings", balance: 234343);
16     }
17 }
```

output



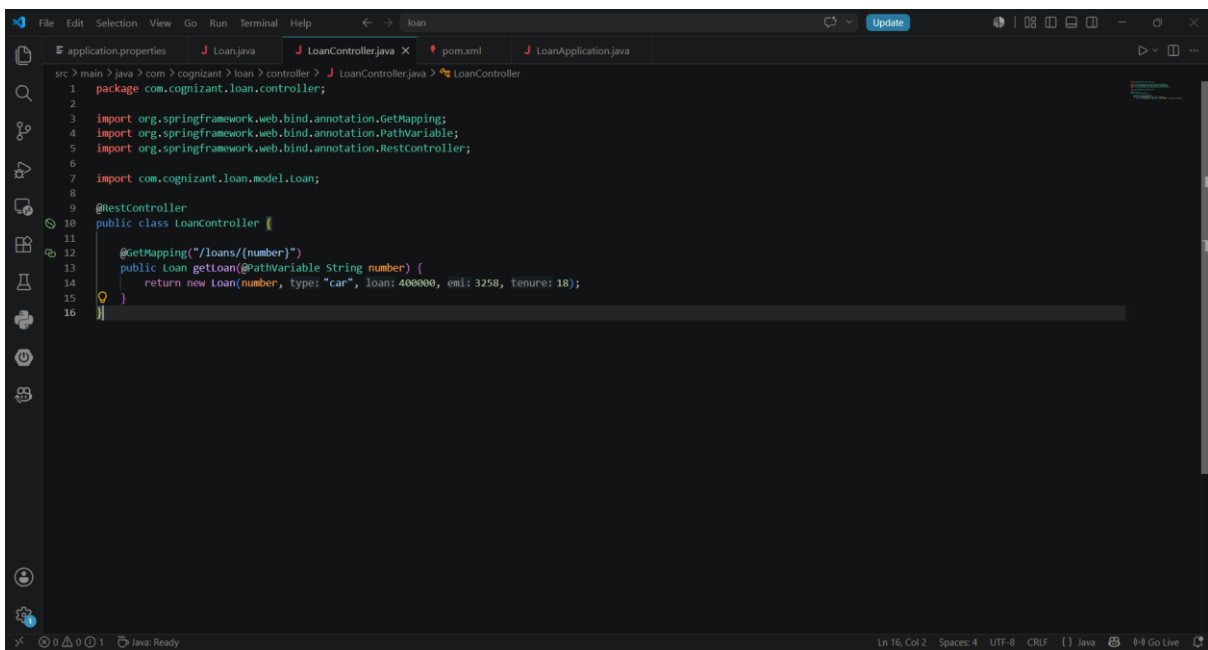
Loan

Loan.java



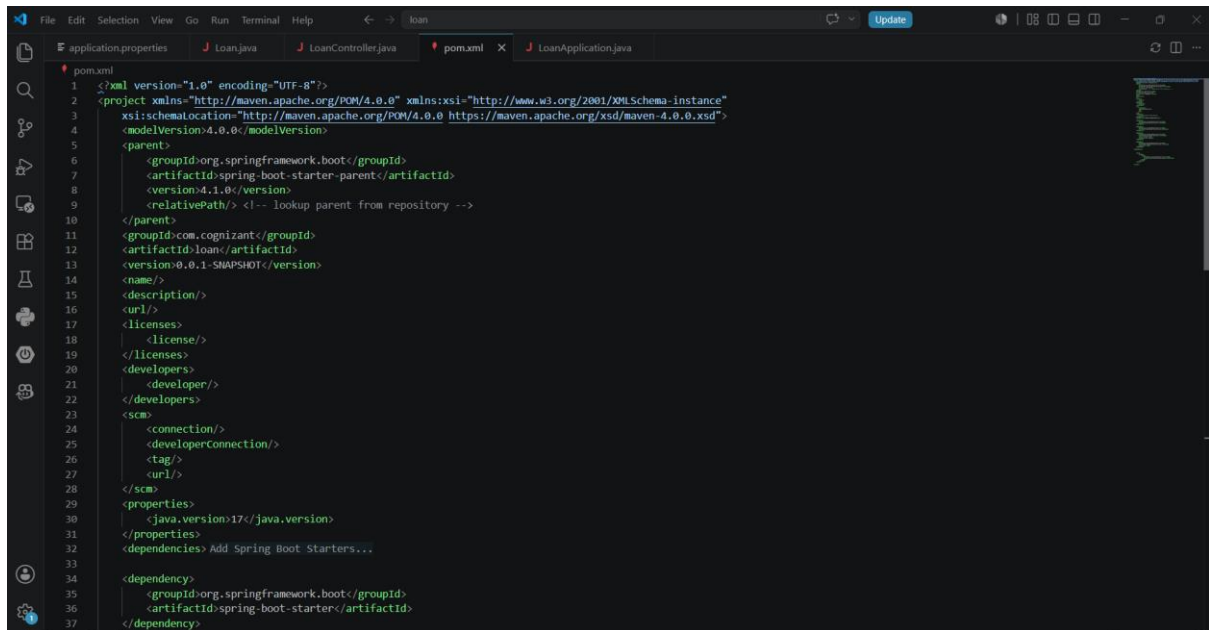
```
1 package com.cognizant.loan.model;
2
3 public class Loan {
4
5     private String number;
6     private String type;
7     private double loan;
8     private int emi;
9     private int tenure;
10
11     public Loan() {
12     }
13
14     public Loan(String number, String type, double loan, int emi, int tenure) {
15         this.number = number;
16         this.type = type;
17         this.loan = loan;
18         this.emi = emi;
19         this.tenure = tenure;
20     }
21
22     public String getNumber() {
23         return number;
24     }
25
26     public void setNumber(String number) {
27         this.number = number;
28     }
29
30     public String getType() {
31         return type;
32     }
33
34     public void setType(String type) {
35         this.type = type;
36     }
37 }
```

Loancontroller.java

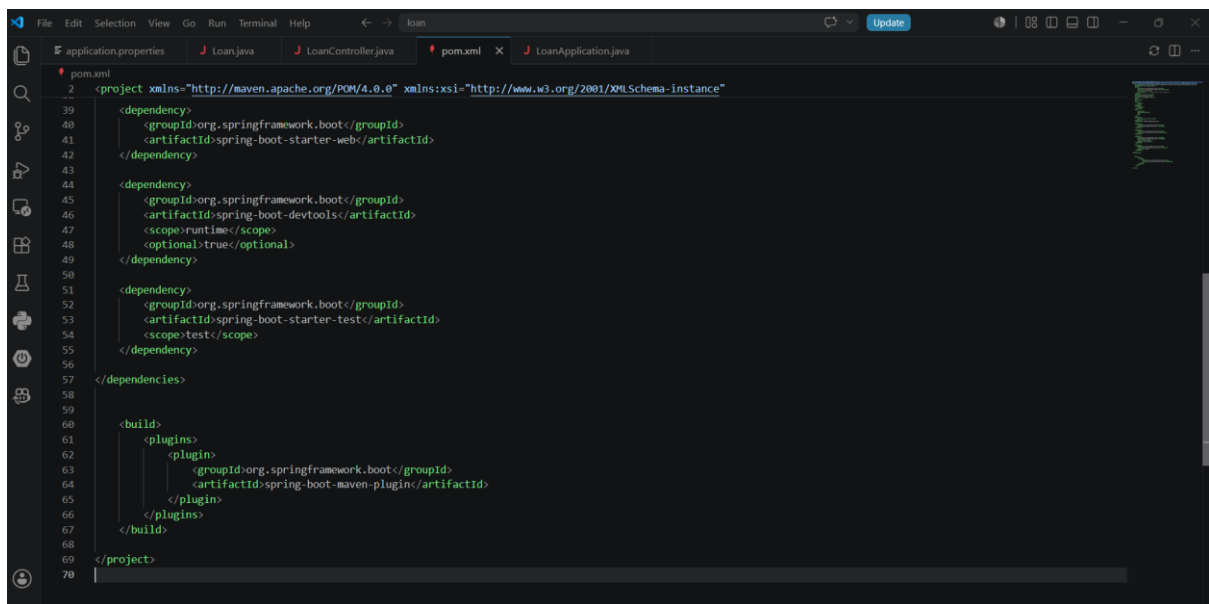


```
1 package com.cognizant.loan.controller;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.PathVariable;
5 import org.springframework.web.bind.annotation.RestController;
6
7 import com.cognizant.loan.model.Loan;
8
9 @RestController
10 public class LoanController {
11
12     @GetMapping("/loans/{number}")
13     public Loan getLoan(@PathVariable String number) {
14         return new Loan(number, type: "car", loan: 400000, emi: 3258, tenure: 18);
15     }
16 }
```

Pom.xml

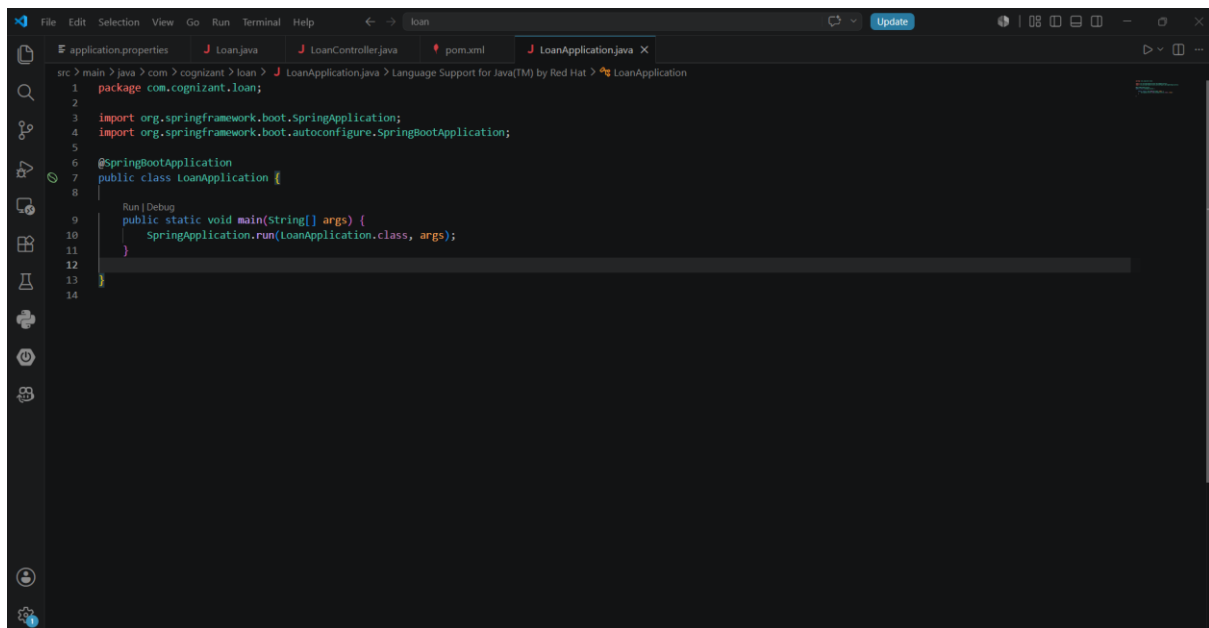


```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
4   <modelVersion>4.0.0</modelVersion>
5   <parent>
6     <groupId>org.springframework.boot</groupId>
7     <artifactId>spring-boot-starter-parent</artifactId>
8     <version>4.1.0</version>
9     <relativePath/> <!-- lookup parent from repository -->
10  </parent>
11  <groupId>com.cognizant</groupId>
12  <artifactId>loans</artifactId>
13  <version>0.0.1-SNAPSHOT</version>
14  <name/>
15  <description/>
16  <url/>
17  <licenses>
18    <license/>
19  </licenses>
20  <developers>
21    <developer/>
22  </developers>
23  <scm>
24    <connection/>
25    <developerConnection/>
26    <tag/>
27    <url/>
28  </scm>
29  <properties>
30    <java.version>17</java.version>
31  </properties>
32  <dependencies> Add Spring Boot Starters...
33
34  <dependency>
35    <groupId>org.springframework.boot</groupId>
36    <artifactId>spring-boot-starter</artifactId>
37  </dependency>
```



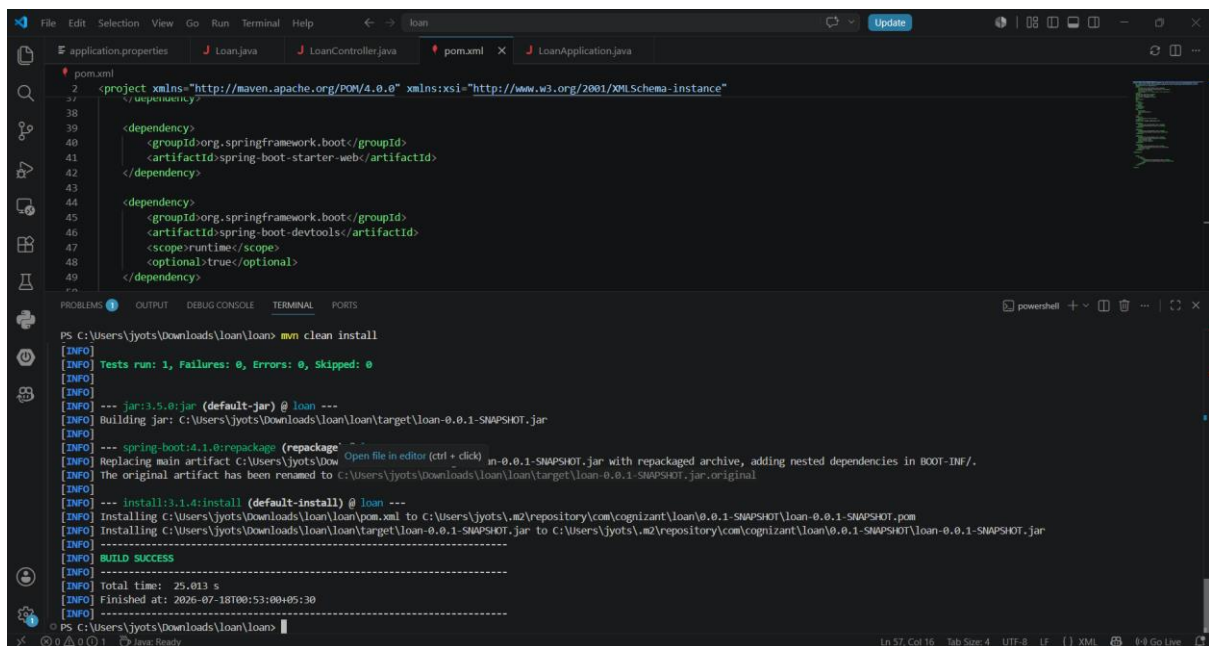
```
2 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
4   <modelVersion>4.0.0</modelVersion>
5   <parent>
6     <groupId>org.springframework.boot</groupId>
7     <artifactId>spring-boot-starter-parent</artifactId>
8     <version>4.1.0</version>
9     <relativePath/> <!-- lookup parent from repository -->
10  </parent>
11  <groupId>com.cognizant</groupId>
12  <artifactId>loans</artifactId>
13  <version>0.0.1-SNAPSHOT</version>
14  <name/>
15  <description/>
16  <url/>
17  <licenses>
18    <license/>
19  </licenses>
20  <developers>
21    <developer/>
22  </developers>
23  <scm>
24    <connection/>
25    <developerConnection/>
26    <tag/>
27    <url/>
28  </scm>
29  <properties>
30    <java.version>17</java.version>
31  </properties>
32  <dependencies>
33    <dependency>
34      <groupId>org.springframework.boot</groupId>
35      <artifactId>spring-boot-starter-web</artifactId>
36    </dependency>
37    <dependency>
38      <groupId>org.springframework.boot</groupId>
39      <artifactId>spring-boot-devtools</artifactId>
40      <scope>runtime</scope>
41      <optional>true</optional>
42    </dependency>
43    <dependency>
44      <groupId>org.springframework.boot</groupId>
45      <artifactId>spring-boot-starter-test</artifactId>
46      <scope>test</scope>
47    </dependency>
48  </dependencies>
49
50  <build>
51    <plugins>
52      <plugin>
53        <groupId>org.springframework.boot</groupId>
54        <artifactId>spring-boot-maven-plugin</artifactId>
55      </plugin>
56    </plugins>
57  </build>
58 </project>
```

LoanApplication.java



```
src > main > java > com > cognizant > loan > LoanApplication.java > Language Support for Java(TM) by Red Hat > LoanApplication
1 package com.cognizant.loan;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6 @SpringBootApplication
7 public class LoanApplication {
8
9     Run | Debug
10     public static void main(String[] args) {
11         SpringApplication.run(LoanApplication.class, args);
12     }
13
14 }
```

Output



```
pom.xml
1 <?xml version="1.0" encoding="UTF-8"?>
2 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3     xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
4     <modelVersion>4.0.0</modelVersion>
5     <groupId>com.cognizant</groupId>
6     <artifactId>loan</artifactId>
7     <version>0.0.1-SNAPSHOT</version>
8     <packaging>jar</packaging>
9
10     <dependencies>
11         <dependency>
12             <groupId>org.springframework.boot</groupId>
13             <artifactId>spring-boot-starter-web</artifactId>
14         </dependency>
15         <dependency>
16             <groupId>org.springframework.boot</groupId>
17             <artifactId>spring-boot-devtools</artifactId>
18             <scope>runtime</scope>
19             <optional>true</optional>
20         </dependency>
21     </dependencies>
22
23     <build>
24         <plugins>
25             <plugin>
26                 <groupId>org.springframework.boot</groupId>
27                 <artifactId>spring-boot-maven-plugin</artifactId>
28             </plugin>
29         </plugins>
30     </build>
31 </project>
```

```
PS C:\Users\jyots\Downloads\loan\loan> mvn clean install
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ loan ---
[INFO] Building jar: C:\Users\jyots\Downloads\loan\loan\target\loan-0.0.1-SNAPSHOT.jar
[INFO]
[INFO] --- spring-boot:4.1.0:repackage (repackage) @ loan ---
[INFO] Replacing main artifact C:\Users\jyots\Downloads\loan\loan\target\loan-0.0.1-SNAPSHOT.jar with repackaged archive, adding nested dependencies in BOOT-INF/.
[INFO] The original artifact has been renamed to C:\Users\jyots\Downloads\loan\loan\target\loan-0.0.1-SNAPSHOT.jar.original
[INFO]
[INFO] --- install:3.1.4:install (default-install) @ loan ---
[INFO] Installing C:\Users\jyots\Downloads\loan\loan\target\loan-0.0.1-SNAPSHOT.jar to C:\Users\jyots\Downloads\loan\loan\repository\com\cognizant\loan\0.0.1-SNAPSHOT\loan-0.0.1-SNAPSHOT.jar
[INFO] Installing C:\Users\jyots\Downloads\loan\loan\target\loan-0.0.1-SNAPSHOT.jar.original to C:\Users\jyots\Downloads\loan\loan\repository\com\cognizant\loan\0.0.1-SNAPSHOT\loan-0.0.1-SNAPSHOT.jar.original
[INFO] BUILD SUCCESS
[INFO] Total time: 25.013 s
[INFO] Finished at: 2026-07-18T00:53:00+05:30
[INFO]
PS C:\Users\jyots\Downloads\loan\loan>
```